TOPIC 1 · FUNDAMENTALS OF MICROSOFT 365 COPILOT
LAB 1 THEME · EXCEL TO WORD

Lab 1 — Prepare Work Orders and Draft the Operations Brief

Maps to	LO1: integrate verified Word, Excel, PowerPoint and Outlook outputs
Duration	30 minutes
Tools	Microsoft 365 Copilot · Excel · Word · PowerPoint · Outlook · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-1

Goal

You are Northstar Water Services' maintenance coordinator. A Q3 export contains repeated IDs and inconsistent category labels. You will clean a copy in Excel, reconcile its controls, and use the verified result to draft a Word operations brief. The same site, status and assignment definitions underpin every later report.

What You Will Do

You are Northstar Water Services' maintenance coordinator. A Q3 export contains repeated IDs and inconsistent category labels. You will clean a copy in Excel, reconcile its controls, and use the verified result to draft a Word operations brief. The same site, status and assignment definitions underpin every later report.

What You Will Build

Northstar_Clean_Work_Orders.xlsx and Northstar_Operations_Brief.docx, with reconciled counts, reporting scope and a source register.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Copy data/Northstar_Work_Orders.xlsx to your OneDrive work-genai folder as Northstar_Clean_Work_Orders.xlsx. Open it in Excel, set Formulas > Calculation Options > Automatic, and read About_This_Data and Control. The workbook is the deliberately dirty 122-row input. Open Copilot from Home > Copilot.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Read About_This_Data and Control for Northstar Water Services. Explain what one water-pipe
service work order represents, the reporting date, the five job categories and how current
owners differ from proposed owners. Do not edit or infer customer impacts.
```

Step 2. Use Chat only to inspect the named table before making changes. Record row count and the two exact repeated IDs. A Work_Order_ID is the row grain; do not count repeated export records twice.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Inspect tblWorkOrders. Report row count, duplicate Work_Order_ID values, category spelling
variants and missing Assigned_To. Do not edit. Treat 2026-09-30 as the fixed reporting date.
```

Step 3. Choose Plan if available. Read the proposed cleaning plan before using Allow editing. The duplicates are exact repeated export records: keep one per ID. Trim and standardise category labels to Leak Repair, Valve Repair, Pipe Replacement, Pump Service, Meter Service; preserve missing owners and all unique source values. Save a copy of the raw sheet before the change.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
Plan to retain an untouched Raw_Export copy, remove only exact duplicate records by
Work_Order_ID, and standardise category spelling in tblWorkOrders. Keep dates, statuses and
assignments. Ask if two records with one ID differ.

EXCEL - ALLOW EDITING:
Apply the reviewed plan. Keep one identical record per repeated ID. Do not guess missing
owners. Reconcile the cleaned table against Control.
```

Step 4. Add a Checks sheet or use the supplied one. Ask for formulas, then inspect the table and formulas yourself. Overdue excludes Closed and Cancelled and uses strictly earlier than the fixed date, so a due-today job is not overdue. Record each verified value in the Evidence_Log sheet (claim, value, source, scope and as-at date).

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
In Checks calculate total rows, closed, cancelled, backlog, overdue backlog, unassigned
backlog and total Cost_SGD for tblWorkOrders. Use Control!B2 as reporting date. Define backlog
as Status neither Closed nor Cancelled. Add the identity total = closed + cancelled + backlog.
Do not change source data.
```

Step 5. Open data/templates/Northstar_Operations_Brief_Template.docx in Word and save as Northstar_Operations_Brief.docx in OneDrive. In Copilot select the verified source file using Reference a file or /. If Word cannot use the workbook, paste your verified evidence table plus the definitions from the Control sheet into the document and select that text; do not imply Word reads the Excel chat.

PROMPT / CHECK — copy and adapt

```
WORD COPILOT:
Draft a one-page Northstar operations brief from the verified evidence supplied here. Include
reporting scope/date, backlog and overdue findings, current assignment issues, proposed next
actions and open questions. Keep observed facts separate from proposals. Preserve numbers and
definitions. Use UNKNOWN for causes, approvals or capacity decisions not evidenced.
```

Step 6. Compare every number and work-order ID in the Word brief to Checks and the evidence register. Add the OneDrive source link and all-filters scope. Log the Word prompt manually in Excel Prompt_Log because Excel cannot access the Word conversation. Save the cleaned workbook and brief for Lab 2.

PROMPT / CHECK — copy and adapt

```
WORD COPILOT:
Audit this Northstar Water Services operations brief against the verified evidence table I
supply. Return a discrepancy list for counts, cost, reporting scope, overdue definition and
assignment claims. Identify missing source links as UNKNOWN. Do not change facts or approve
work.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-1

#	Purpose	Prompt
1	Scope	State actual source, report date and selected scope before acting.
2	Verify	Explain the source/formula checks and preserve unknowns and original records.

Test It

Check	Expected result
Clean table	120 unique Work_Order_ID rows; five canonical job categories.
Portfolio identity	120 = 48 Closed + 12 Cancelled + 60 backlog.
Evidence handoff	Overdue 50; unassigned backlog 6; total cost S$35,827.50. Word agrees with Excel.
No guessed owner	Blank assignments remain blank; proposals labelled unapproved.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?